

Day 13 Problem Set

Geometry Summer Credit | Sine, Cosine, and Tangent Ratios

Name:	Date:	Period:
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Goal

Practice labeling sides, choosing the correct trig ratio, and using calculators carefully in degree mode.

Support Practice: Label and Choose

1.	From angle A, name the ratio that uses opposite and hypotenuse.
2.	From angle A, name the ratio that uses adjacent and hypotenuse.
3.	From angle A, name the ratio that uses opposite and adjacent.
4.	A side is across from the right angle. What is it called?
5.	Write the setup to find an opposite side when angle A and the hypotenuse are known.

Core Practice: Find Missing Sides

6.	A right triangle has angle 30 degrees and hypotenuse 18. Find the opposite side.
7.	A right triangle has angle 40 degrees and hypotenuse 25. Find the adjacent side.
8.	A right triangle has angle 55 degrees and adjacent side 14. Find the opposite side.

9. A right triangle has angle 22 degrees and opposite side 8. Find the hypotenuse.

10. A right triangle has angle 68 degrees and adjacent side 5. Find the hypotenuse.

11. A ramp makes a 12 degree angle with the ground. The ramp is 18 ft long. How high does it rise?

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Continued

Core Practice: Find Missing Angles and Review

12. A right triangle has opposite side 9 and hypotenuse 20 from angle A. Find angle A.

13. A right triangle has adjacent side 16 and hypotenuse 25 from angle B. Find angle B.

14. A right triangle has opposite side 11 and adjacent side 7 from angle C. Find angle C.

15. A ladder is 20 ft long and reaches 16 ft up a wall. Find the angle the ladder makes with the ground.

16. A kite string is 45 m long and the kite is 30 m above the ground. Find the angle of elevation.

17. Spiral review: A 45-45-90 triangle has hypotenuse $9\sqrt{2}$. Find each leg.

18. Spiral review: A 30-60-90 triangle has hypotenuse 24. Find both legs.

19. Spiral review: Find the distance between $(-1, -2)$ and $(5, 6)$.

20. Challenge: Explain why you cannot choose sine, cosine, or tangent until you know the reference angle.